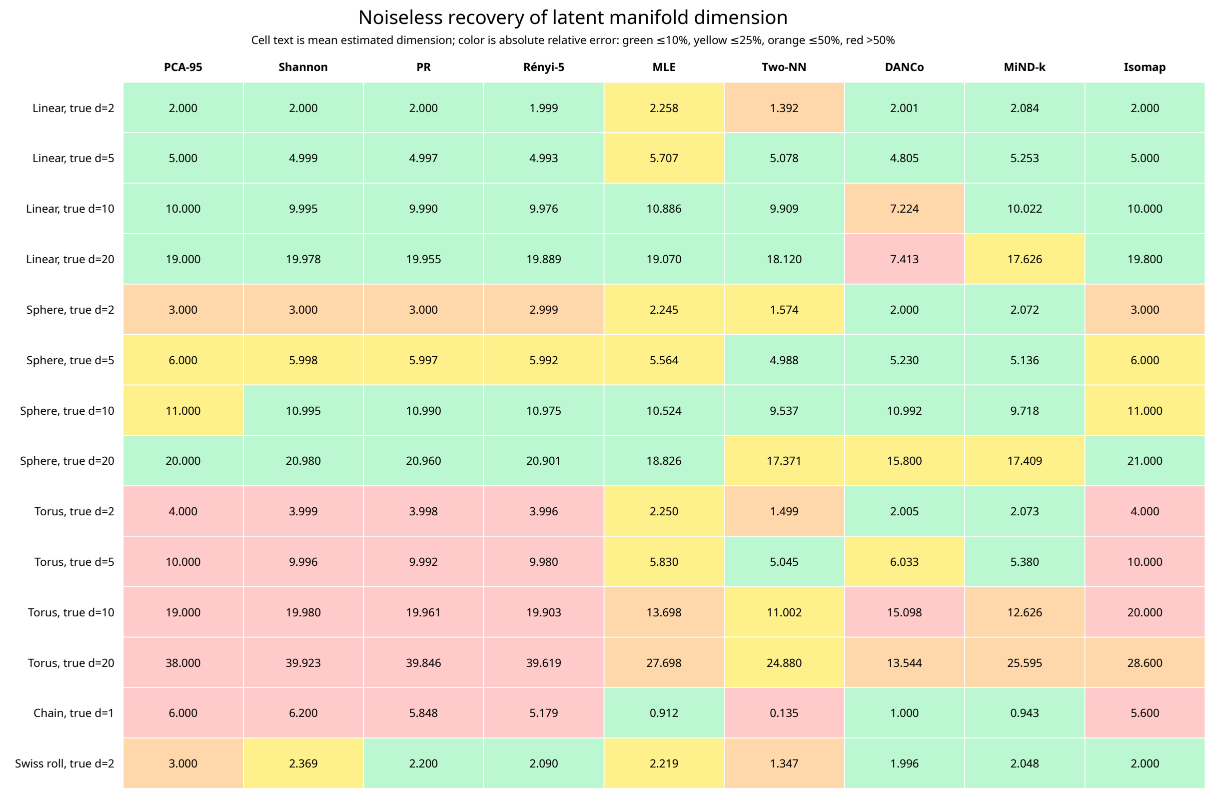
EffDim recovery on noisy embedded manifolds

Linear subspaces, spheres, tori, a nonlinear chain, and a Swiss roll were sampled with 10,000 points, randomly embedded into 256 dimensions, and evaluated over five replicates. Core estimators used exact GPU neighbours; Landmark Isomap used a CAGRA `k=10` graph with 512 landmarks.

Main findings

1. **On clean manifolds, MiND-MLk is best overall by median relative error (4.6%), followed by DANCo (7.3%), Two-NN (11.6%), and MLE/TLE (11.8%).** No local method is uniformly best: DANCo saturates on higher dimensions, while Two-NN badly underestimates the nonlinear chain.
2. **Landmark Isomap exactly recovers clean linear subspaces and the Swiss roll, but its median clean error is 15%.** Closed topology prevents a globally isometric Euclidean unfolding: spheres return about d+1 and tori about 2d. The nonlinear chain's `k=10` graph retains only about 46% of points in its largest component, invalidating that estimate.
3. **Spectral metrics measure global embedding span, not nonlinear latent dimension.** A sphere of intrinsic dimension d spans d+1 linear axes; a d-torus spans 2d; and the one-dimensional chain has PCA-95 of 6.
4. **At 30 dB, Two-NN has the lowest selected-method median error (12.2%),** followed by participation ratio (15.1%) and MiND-MLk (19.5%). Local neighbourhood geometry is already measurably distorted.
5. **At 20 dB, participation ratio is best by median error (17.3%).** Isomap follows at 22.5%; selected local estimators rise to approximately 30–47%.
6. **At 10 dB, none of the methods reliably recovers the clean latent dimension.** Gaussian ambient noise makes support dimension 256; PCA-95 is particularly inflated, with median error around 1,890%.
7. **There is no universal winner.** MiND-MLk is strongest for clean manifold recovery, participation ratio is comparatively robust to moderate noise, Isomap is useful for unfoldable connected manifolds, and PCA-95 remains a global variance/compression dimension.
8. **On the real embeddings, Landmark Isomap is stable from k=10 to k=20.** JWST estimates are 9.6, 9.4, and 9.0; DESI 6.6, 6.2, and 6.2; and Legacy Survey 7.8, 8.4, and 8.2. Every graph remains fully connected, and each dataset spans at most 0.6 dimensions across neighbourhood sizes. These values align more closely with participation ratio and higher-order Rényi dimensions than PCA-95, but no ground truth is available.

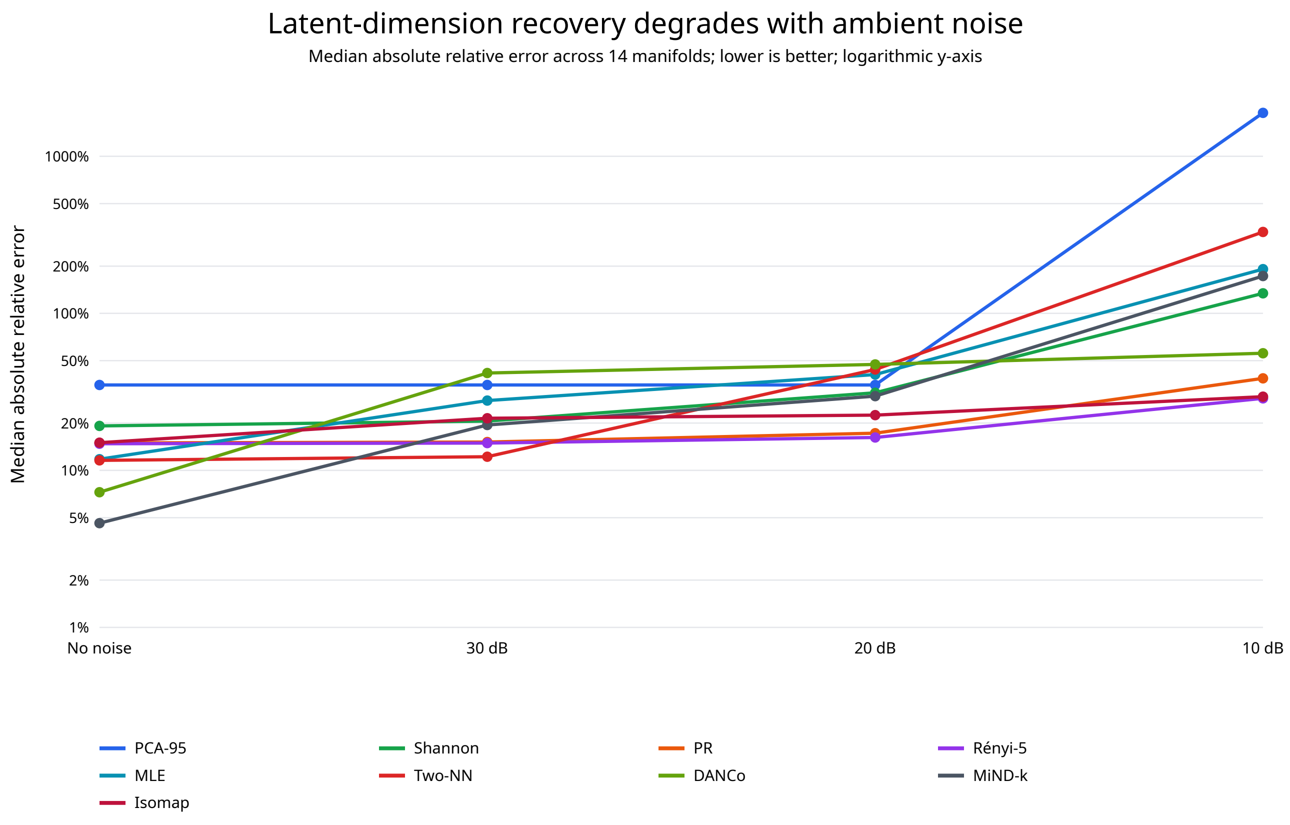
Clean-manifold recovery matrix


Noiseless recovery of latent manifold dimension

Cell text is mean estimated dimension; color is absolute relative error: green ≤10%, yellow ≤25%, orange ≤50%, red >50%

	PCA-95	Shannon	PR	Rényi-5	MLE	Two-NN	DANCo	MiND-k	Isomap
Linear, true d=2	2.000	2.000	2.000	1.999	2.258	1.392	2.001	2.084	2.000
Linear, true d=5	5.000	4.999	4.997	4.993	5.707	5.078	4.805	5.253	5.000
Linear, true d=10	10.000	9.995	9.990	9.976	10.886	9.909	7.224	10.022	10.000
Linear, true d=20	19.000	19.978	19.955	19.889	19.070	18.120	7.413	17.626	19.800
Sphere, true d=2	3.000	3.000	3.000	2.999	2.245	1.574	2.000	2.072	3.000
Sphere, true d=5	6.000	5.998	5.997	5.992	5.564	4.988	5.230	5.136	6.000
Sphere, true d=10	11.000	10.995	10.990	10.975	10.524	9.537	10.992	9.718	11.000
Sphere, true d=20	20.000	20.980	20.960	20.901	18.826	17.371	15.800	17.409	21.000
Torus, true d=2	4.000	3.999	3.998	3.996	2.250	1.499	2.005	2.073	4.000
Torus, true d=5	10.000	9.996	9.992	9.980	5.830	5.045	6.033	5.380	10.000
Torus, true d=10	19.000	19.980	19.961	19.903	13.698	11.002	15.098	12.626	20.000
Torus, true d=20	38.000	39.923	39.846	39.619	27.698	24.880	13.544	25.595	28.600
Chain, true d=1	6.000	6.200	5.848	5.179	0.912	0.135	1.000	0.943	5.600
Swiss roll, true d=2	3.000	2.369	2.200	2.090	2.219	1.347	1.996	2.048	2.000

Noise sensitivity



Median absolute relative error across all shapes

Method	No noise	30 dB	20 dB	10 dB
PCA-95	35.0%	35.0%	35.0%	1890.0%
Participation ratio	15.0%	15.1%	17.3%	38.6%
Shannon ED	19.2%	20.6%	31.2%	134.3%
Rényi α=2	15.0%	15.1%	17.3%	38.6%
Rényi α=3	14.9%	15.0%	16.5%	31.8%
Rényi α=4	14.8%	15.0%	16.3%	29.8%
Rényi α=5	14.8%	14.9%	16.2%	28.7%
Geometric mean ED	8807103441170767.0%	14620.8%	1494.4%	86.5%
MLE	11.8%	27.8%	40.7%	191.2%
Two-NN	11.6%	12.2%	43.9%	329.7%
DANCo	7.3%	41.7%	47.3%	55.7%
MiND-MLi	68.2%	66.0%	77.5%	69.9%
MiND-MLk	4.6%	19.5%	29.7%	172.7%
ESS	87.3%	88.4%	91.4%	95.3%
TLE	11.8%	27.8%	40.7%	191.2%
Landmark Isomap	15.0%	21.5%	22.5%	29.5%

Mean noiseless estimates

Shape	True d	PCA-95	Shannon	PR	Rényi-5	MLE	Two-NN	DANCo	MiND-k	Isomap
Linear	2	2.000	2.000	2.000	1.999	2.258	1.392	2.001	2.084	2.000
Linear	5	5.000	4.999	4.997	4.993	5.707	5.078	4.805	5.253	5.000
Linear	10	10.000	9.995	9.990	9.976	10.886	9.909	7.224	10.022	10.000
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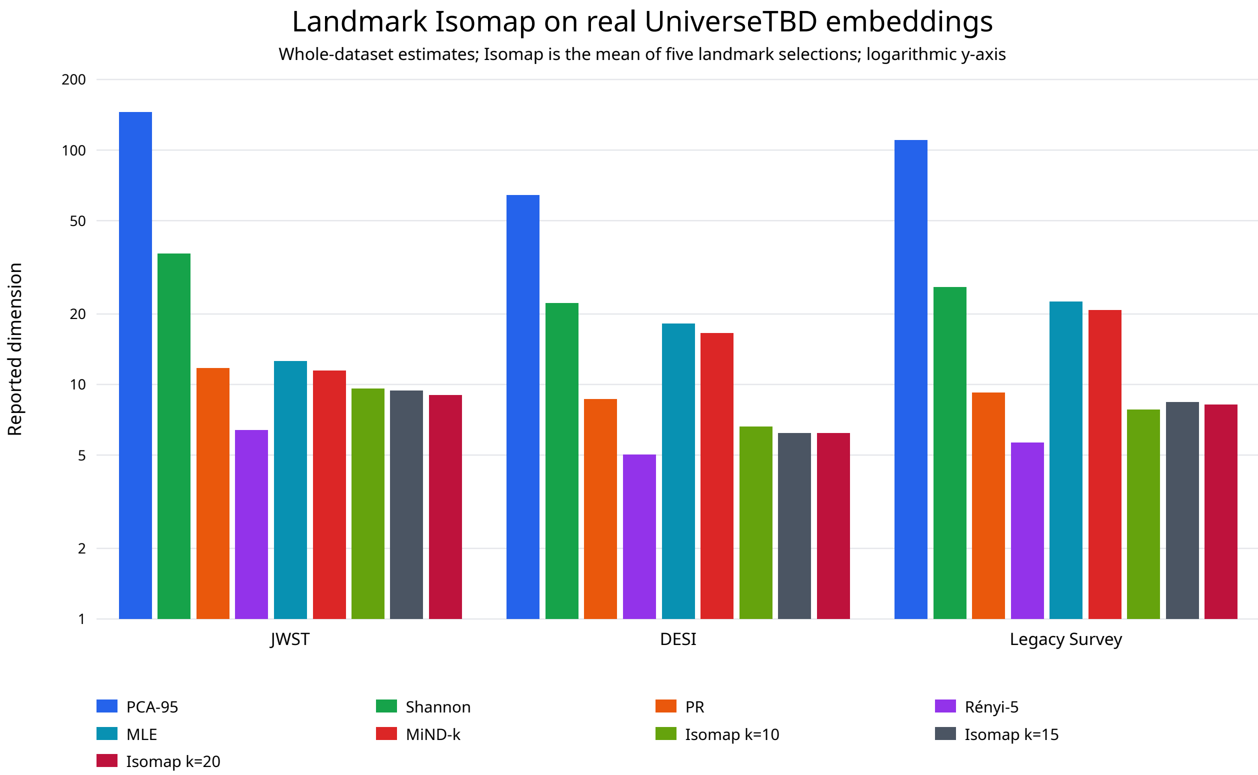
Landmark Isomap on real embeddings



All CAGRA `k=10`, `k=15`, and `k=20` graphs formed one connected component containing 100% of the data. Isomap entries are mean ± SD across five independent landmark selections. Other methods are deterministic.

Method	JWST	DESI	Legacy Survey
PCA-95	145.000	64.000	110.000
Shannon ED	36.134	22.157	25.981
Participation ratio	11.752	8.659	9.241
Rényi α=5	6.388	5.027	5.633
MLE	12.560	18.172	22.560
MiND-MLk	11.464	16.539	20.712
Landmark Isomap k=10	9.6 ± 0.9	6.6 ± 0.9	7.8 ± 0.8
Landmark Isomap k=15	9.4 ± 0.5	6.2 ± 0.4	8.4 ± 1.1
Landmark Isomap k=20	9.0 ± 0.0	6.2 ± 0.4	8.2 ± 1.1

The accompanying CSV contains mean, standard deviation, and relative error for all 15 non-GMST methods plus Isomap under every synthetic shape and noise condition.